## Project Report on

# AIR POLLUTION MANAGEMENT SYSTEM

GD GOENKA SCHOOL | EAST DELHI
Thrive. For Life.


**Submitted By**

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# CERTIFICATE

This is to certify that Anjini Agarwal of Class XII B has prepared the report on the Project entitled "Air Pollution Management System". The report is the result of her efforts & endeavors. The report is found worthy of acceptance as final project report for the subject Computer Science of Class XII. She has prepared the report under my guidance.

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# DECLARATION

I hereby declare that the project work entitled "Air Pollution Management System", submitted to Department of Computer Science, G.D. Goenka Public School, East Delhi is prepared by me. The project work is result of my personal efforts.

Anjini Agarwal
Class: XII B

# ACKNOWLEDGEMENT

I would like to express a deep sense of thanks & gratitude to my project guide Ms. Somya Gupta for guiding me immensely through the course of the project. She always evinced keen interest in my work. Her constructive advice & constant motivation has been responsible for the successful completion of this project.

Anjini Agarwal
Class: XII B

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# MODULES IMPORTED

The project Air Pollution Management System uses three modules for the functionality of the project which are as follows: -

**1. pandas:**

Pandas is an open-source library in Python that is made mainly for working with relational or labeled data both easily and intuitively. It provides various data structures and operations for manipulating numerical data and time series. This library is built on top of the NumPy library of Python. Pandas is fast and it has high performance & productivity for users.

**2. matplotlib:**

Matplotlib is an amazing visualization library in Python for 2D plots of arrays. Matplotlib is a multi-platform data visualization library built on NumPy arrays and designed to work with the broader SciPy stack. It was introduced by John Hunter in the year 2002. One of the greatest benefits of visualization is that it allows us visual access to huge amounts of data in easily digestible visuals. Matplotlib consists of several plots like line, bar, scatter, histogram etc.

**3. sys:**

The sys module in Python provides various functions and variables that are used to manipulate different parts of the Python runtime environment. It allows operating on the interpreter as it provides access to the variables and functions that interact strongly with the interpreter.

# WORKING DESCRIPTION

The Project entitled Air Pollution Management System consists of three built-in modules as stated above and has been used together. The project monitors and represents the data of Air pollution and levels of NO2, SO2 and RSPM in various locations of Delhi as specified in the CSV file. The project also maintains a CSV file to represent the record of location wise NO2, SO2 and RSPM levels.

The project has one user-defined module which is based on the menu-driven approach to perform various manipulations on the csv and data.

The Main Menu of the project further has four submenus:

1- Data Visualization Menu which represents the data through line and bar charts and plots the levels of NO2, SO2 and RSPM at different locations respectively.

2- Data Manipulation Menu which manipulates the data by either adding or deleting records or deleting a specific parameter of air in different locations.

3-Data Analysis Menu contains many options which retrieves data like printing top or bottom records, printing specific or multiple columns or information and displaying statistics as well as information of the data set.

4-Exit: To exit the System. And terminate the program.

The functions used in the project are described as follows:

1. **Menuset():**

   This function displays the various menu options to the user so that the required choice can be entered and the functions can be performed accordingly.

2. **Display():**

   This function represents the data visualization menu and graphically displays data through line chart and bar chart based on the user's choice. The function

allows the user to represent data of $SO_2$ and $NO_2$ of specific location through line chart and bar chart.

**3. <u>Manipulation()</u>**

This function displays the various sub-options available in the data manipulation menu. It allows the user to enter his choice and perform various operations such as adding area-wise air quality parameters, deleting details of air parameters and deleting specific air parameters.

**4. <u>Analysis()</u>**

This function displays the various sub-options available in the data analysis menu. It allows the user to enter his choice and perform various operations on the data frame such as displaying specific records from top and bottom, displaying specific single or multiple columns, displaying station details based on the high $NO_2$ and low rspm levels. The function also displays complete statistics and complete information about the dataframe.

**5. <u>Bar_chart()</u>**

This function reads the data frame and retrieves only location column from it to perform the required operation. This function displays date-wise $SO_2$ and $NO_2$ readings of a specific location through bar chart.

**6. <u>Bar_chart2()</u>**

This function reads the data frame and retrieves only location column from it to perform the required operation. This function displays date-wise rspm readings of a specific location through bar chart.

**7. <u>Line_chart()</u>**

This function reads the data frame and retrieves only location column from it to perform the required operation. This function displays date-wise $SO_2$ and $NO_2$ readings of a specific location through line chart.

**8. <u>Line_chart2()</u>**

This function reads the data frame and retrieves only location column from it to perform the required operation. This function displays date-wise $SO_2$ and $NO_2$ readings of a specific location through line chart.

**9. <u>Exit()</u>**

This function allows the user to exit the project once all options have been successfully executed.

# PROJECT CODE

```
#-----------------------------------------------------------------------------------------------------
# Imported Libraries and CSV Files
#-----------------------------------------------------------------------------------------------------
import pandas as pd
import numpy as np
import sys
import matplotlib.pyplot as plt

df=pd.read_csv("Updated CSV1-copy.csv")
locationcsv=pd.read_csv("UPLocations.csv")
print(df)

#-----------------------------------------------------------------------------------------------------
#Functions For Displaying The Main Menu
#-----------------------------------------------------------------------------------------------------
def Menuset():
    ans='y'
    while ans=='y' or ans=='Y':
        print("\t\t==============================")
        print("\t\t   Air Polloution in Delhi   ")
        print("\t\t==============================")
        print("\t\t1: Data Manipulation Menu\n")
        print("\t\t2: Data Analysis Menu\n")
        print("\t\t3: Data Visualization Menu\n")
        print("\t\t4: Exit Menu\n")
        print("\t\t==============================")
        opt=input("\t\tEnter your choice: ")
        if opt=='1':
            manipulation()
        elif opt=='2':
            analysis()
        elif opt=='3':
            display()
```

```
        elif opt=='4':
            chance=input("Are You Sure to exit(Y/N)")
            print('THANK You. Closing now....')
            sys.exit()
        else :
            print('Invalid option. Try Again')
            continue

#-----------------------------------------------------------------------------------------------------------
# For data manipulation
#-----------------------------------------------------------------------------------------------------------

def manipulation():
    while True:
        global df1
        df1=pd.DataFrame()
        print("\t\tManipulation Menu")
        print("\t\t*****************")
        print("""\t\t1. Insert Details of Area-wise Air Quality parameters\n
                \t\t2. Delete Details\n
                \t\t3. Delete a specific air quality parameter\n
                \t\t4. Go back to main menu\n""")
        x2=int(input("\t\tEnter your choice: "))
        if x2==1:
            col=df.columns
            print(col)
            print(df.head(1))
            p=0
            lst1=[]
            lst1=eval(input('Enter a list of values according to the columns: '))
            print(lst1)
            '''s1=pd.Series(lst1)
            print(s1)
            s1=s1.T
            print(s1)'''
            #df1=df.append(s1,ignore_index=True)
            '''df1=df.append(lst1,ignore_index=True)'''
```

```
            count=len(df.index.values)
            df.loc[count,:]=lst1
            print('Insertion Completed !!!!')
            print(df)
        elif x2==2:
            print(df)
            x3=int(input('Enter the row index you want to delete: '))
            df1=df.drop(x3)
            print('Row No.',x3+1,' is deleted')
            print(df1)
        elif x2==3:
            print(df.columns)
            p2=input('Enter the Column Name you want to delete: ')
            w1=input('Are you Sure to delete(Y,N)')
            if w1=='y' or w1=='Y':
                del df[p2]
                print('Column:',p2,'has been deleted successfully')
                df2=pd.DataFrame()
                df2=df
                print(df2)
        elif x2==4:
            break

#-----------------------------------------------------------------------------------------------------------
# For Analysis of Data
#-----------------------------------------------------------------------------------------------------------
def analysis():
    while True:
        print("\t\tData Frame Analysis")
        print("\t\t~~~~~~~~~~~~~~~~~~~~~~~~~~~~")
        menu=''' \t\t1. Top record
                \n\t\t 2. Bottom Records
                \n\t\t 3. To Print Specific column
                \n\t\t 4. To Print multiple columns
                \n\t\t 5. To display station details according to highest NO2 content
                \n\t\t 6. To display station details according to lowest rspm parameter
                \n\t\t 7. To Display complete statitics of the dataframe
```

```
        \n\t\t 8. To Display complete information about dataframe
        \n\t\t 9.To go back'''
print(menu)
opt1=int(input('\t\tEnter Your Choice : '))
if opt1==1:
    p3=int(input('Enter the No. of records you want to see: '))
    print('Top',opt1,'records')
    print(df.head(p3))
elif opt1==2:
    p4=int(input('Enter the No. of records you want to see: '))
    print('Bottom',opt1,'records')
    print(df.tail(p4))
elif opt1==3:
    print('Name of column\n',df.columns)
    p5=input('Enter the Column name to be displayed: ')
    print(df[p5])
elif opt1==4:
    print('Name of the columns\n',df.columns)
    p6=eval(input('Enter the column names as list in square brackets[]: '))
    print(df[p6])
elif opt1==5:
    print('Original Dataframe is')
    print(df)
    df1=df.sort_values(by='no2', ascending=False)
    print('station details according to highest NO2 content is')
    print(df1)
elif opt1==6:
    print('Original Dataframe is')
    print(df)
    df1=df.sort_values(by='rspm', ascending=True)
    print('station details according to highest NO2 content is')
    print(df1)
elif opt1==7:
    print('Statistics of the database: ')
    print(df.describe())

  elif opt1==8:
```

```
        print('Information About The Database: ')
        print(df.info())
    elif opt1==9:
         break

#-----------------------------------------------------------------------------------------------------------
# For displaying Data Visualisation Menu
#-----------------------------------------------------------------------------------------------------------
def display():

  while True:
    print()
    print("\t\t=========================")
    print("\t\t Data Visualisation Menu ")
    print("\t\t=========================")
    print("\t\t1: Line Chart Representation of SO2 and NO2 content of specific
      station")
    print("\t\t2: Line chart Representation of RSPM of specific location")
    print("\t\t3: Bar chart Representation of SO2 and NO2 of specific station ")
    print("\t\t4: Bar Chart Representation of RSPM of specific location")
    print("\t\t5: Back")
    opt1=input("\t\tEnter your choice: ")
    if opt1=='1':
        line_chart()
    elif opt1=='2':
        Line_chart2()
    elif opt1=='3':
        Bar_chart()
    elif opt1=='4':
        Bar_chart2()
    elif opt1=='5':
        print("Going Back to Main Menu....")
        break
    else:
        print("Invalid input Try again")

#-----------------------------------------------------------------------------------------------------------
# Displaying Bar chart of RSPM
```

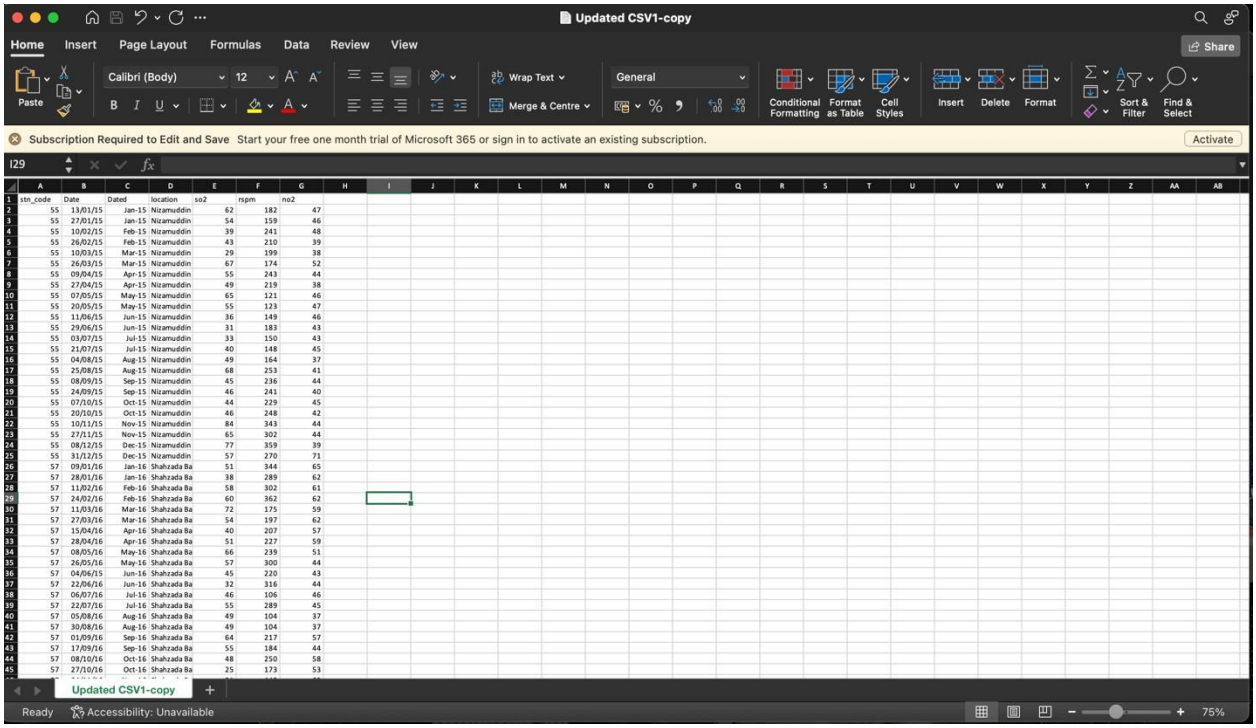
```
#-----------------------------------------------------------------------------------------------------------
def Bar_chart2():
    print("Available Locations In New Delhi")
    h=locationcsv['location']
    print(h)
    V=df.groupby('location')
    cname=input("Enter Location Name: ")
    df1=df.loc[(df['location']==cname)]
    x=np.arange(len(df1))
    if df1.empty!=True:
        dt=df1['Date']
        gas3=df1['rspm']
        plt.bar(x,gas3,label='RSPM',width=0.75,color='m')
        plt.xticks(x,dt,fontsize=6,rotation=90)
        plt.title('AIR POLLOUTION IN DELHI\nLevel Of
Components\n',color='m',fontsize=13)
        plt.xlabel('Date of Reading->',color='m')
        plt.ylabel('RSPM ->>',color='m',fontsize=12)
        plt.grid()
        plt.legend()
        plt.show()

#-----------------------------------------------------------------------------------------------------------
# For displaying Bar Chart of So2 and No2
#-----------------------------------------------------------------------------------------------------------
def Bar_chart():
    print("Available Locations In New Delhi")
    h=locationcsv['location']
    print(h)
    V=df.groupby('location')
    cname=input("Enter Location Name: ")
    df1=df.loc[(df['location']==cname)]
    x=np.arange(len(df1))
    if df1.empty!=True:
        dt=df1['Date']
        gas1=df1['so2']
        gas2=df1['no2']
```

```
        plt.bar(x-0.25,gas1,label='SO2',width=0.5,color='r')
        plt.bar(x+0.25,gas2,label='NO2',width=0.5,color='g')
        plt.xticks(x,dt,fontsize=6,rotation=90)
        plt.title('AIR POLLOUTION IN DELHI\nLevel Of
Components\n',color='blue',fontsize=12)
        plt.xlabel('Date of Reading->',color='g')
        plt.ylabel('SO2 and NO2->',color='g',fontsize=12)
        plt.legend()
        plt.grid()
        plt.show()

#-----------------------------------------------------------------------------------------------------------
# For displaying Line chart of So2 and No2
#-----------------------------------------------------------------------------------------------------------
def line_chart():
    while True:
        X=[]
        V=df.groupby('location')
        print("Available Locations In New Delhi")
        h=locationcsv['location']
        print(h)
        cname=input("Enter Location Name: ")
        df1=df.loc[(df['location']==cname)]
        if df1.empty!=True:
            dt=df1['Date']
            l=np.arange(len(dt))
            y1=df1['so2']
            y2=df1['no2']
            plt.plot(l,y1,label="Level OF SO2",linestyle='solid',marker='^')
            plt.plot(l,y2,label="Level OF NO2",linestyle='solid',marker='o')
            plt.xticks(l,dt,fontsize=6,rotation=90)
            plt.xlabel("Date of Reading->",color='r')
            plt.ylabel("SO2 and NO2->",color='r',fontsize=12)
            plt.title('AIR POLLOUTION IN DELHI\nLevel Of Components\n' + cname,
           color='blue', fontsize=12)
            plt.grid()
            plt.legend()
```

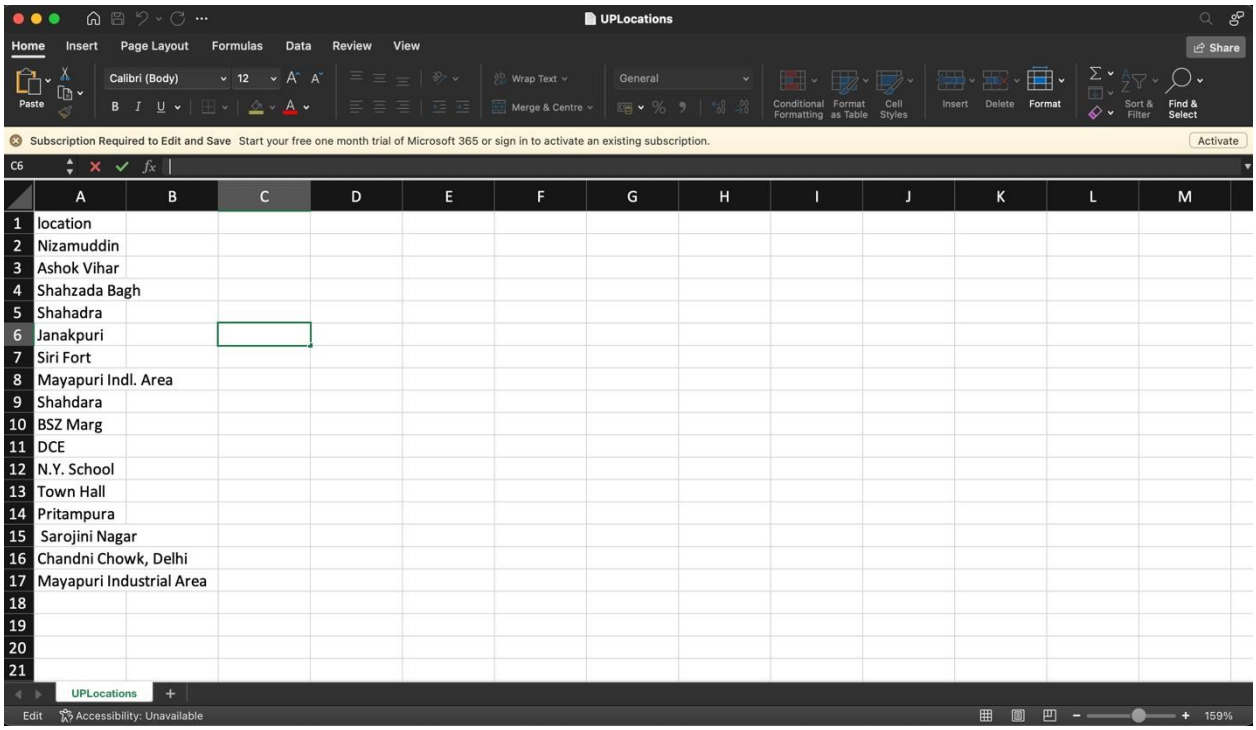
```
            plt.show()
            break
        else:
            print("City Name is incorrect.")
            print("Please Check again and retry")
#-------------------------------------------------------------------------------------------------------
# For displaying Line Chart of RSPM
#-------------------------------------------------------------------------------------------------------
def Line_chart2():
    while True:
        X=[]
        V=df.groupby('location')
        print("Available Locations In New Delhi")
        h=locationcsv['location']
        print(h)
        cname=input("Enter Location Name: ")
        df1=df.loc[(df['location']==cname)]
        if df1.empty!=True:
            dt=df1['Date']
            l=np.arange(len(dt))
            y3=df1['rspm']
            plt.plot(l,y3,label="Level OF RSPM",linestyle='solid',marker='^')
            plt.xticks(l,dt,fontsize=7,rotation=90)
            plt.title('AIR POLLOUTION IN DELHI\n Level Of Components\n'+ cname,
          color='blue',fontsize=12)
            plt.xlabel("Date of Reading->",color='r')
            plt.ylabel("RSPM->",color='r',fontsize=12)
            plt.legend()
            plt.grid()
            plt.show()
            break
        else:
            print("City Name is incorrect.")
            print("Please Check again and retry
Menuset()
#-------------------------------------------------------------------------------------------------------
```

# OUTPUT SCREENS

## CSV files



| stn_code | Date | Dated | location | so2 | rspm | no2 |
|---|---|---|---|---|---|---|
| 55 | 13/01/15 | Jan-15 | Nizamuddin | 62 | 182 | 47 |
| 55 | 27/01/15 | Jan-15 | Nizamuddin | 54 | 159 | 46 |
| 55 | 10/02/15 | Feb-15 | Nizamuddin | 39 | 241 | 48 |
| 55 | 26/02/15 | Feb-15 | Nizamuddin | 43 | 210 | 39 |
| 55 | 10/03/15 | Mar-15 | Nizamuddin | 29 | 199 | 38 |
| 55 | 26/03/15 | Mar-15 | Nizamuddin | 67 | 174 | 52 |
| 55 | 09/04/15 | Apr-15 | Nizamuddin | 55 | 243 | 44 |
| 55 | 27/04/15 | Apr-15 | Nizamuddin | 49 | 219 | 38 |
| 55 | 07/05/15 | May-15 | Nizamuddin | 65 | 121 | 46 |
| 55 | 20/05/15 | May-15 | Nizamuddin | 55 | 123 | 47 |
| 55 | 11/06/15 | Jun-15 | Nizamuddin | 36 | 149 | 46 |
| 55 | 29/06/15 | Jun-15 | Nizamuddin | 31 | 183 | 43 |
| 55 | 03/07/15 | Jul-15 | Nizamuddin | 33 | 150 | 43 |
| 55 | 21/07/15 | Jul-15 | Nizamuddin | 40 | 148 | 45 |
| 55 | 04/08/15 | Aug-15 | Nizamuddin | 49 | 164 | 37 |
| 55 | 25/08/15 | Aug-15 | Nizamuddin | 68 | 253 | 41 |
| 55 | 08/09/15 | Sep-15 | Nizamuddin | 45 | 236 | 44 |
| 55 | 24/09/15 | Sep-15 | Nizamuddin | 46 | 241 | 40 |
| 55 | 07/10/15 | Oct-15 | Nizamuddin | 44 | 229 | 45 |
| 55 | 20/10/15 | Oct-15 | Nizamuddin | 46 | 248 | 42 |
| 55 | 10/11/15 | Nov-15 | Nizamuddin | 84 | 343 | 44 |
| 55 | 27/11/15 | Nov-15 | Nizamuddin | 65 | 302 | 44 |
| 55 | 08/12/15 | Dec-15 | Nizamuddin | 77 | 359 | 39 |
| 55 | 31/12/15 | Dec-15 | Nizamuddin | 57 | 270 | 71 |
| 57 | 09/01/16 | Jan-16 | Shahzada Ba | 51 | 344 | 65 |
| 57 | 28/01/16 | Jan-16 | Shahzada Ba | 38 | 289 | 62 |
| 57 | 11/02/16 | Feb-16 | Shahzada Ba | 58 | 302 | 61 |
| 57 | 24/02/16 | Feb-16 | Shahzada Ba | 60 | 362 | 62 |
| 57 | 11/03/16 | Mar-16 | Shahzada Ba | 72 | 175 | 59 |
| 57 | 27/03/16 | Mar-16 | Shahzada Ba | 54 | 197 | 62 |
| 57 | 15/04/16 | Apr-16 | Shahzada Ba | 40 | 207 | 57 |
| 57 | 28/04/16 | Apr-16 | Shahzada Ba | 51 | 227 | 59 |
| 57 | 08/05/16 | May-16 | Shahzada Ba | 66 | 239 | 51 |
| 57 | 26/05/16 | May-16 | Shahzada Ba | 57 | 300 | 44 |
| 57 | 04/06/15 | Jun-16 | Shahzada Ba | 45 | 220 | 43 |
| 57 | 22/06/16 | Jun-16 | Shahzada Ba | 32 | 316 | 44 |
| 57 | 06/07/16 | Jul-16 | Shahzada Ba | 46 | 106 | 46 |
| 57 | 22/07/16 | Jul-16 | Shahzada Ba | 55 | 289 | 45 |
| 57 | 05/08/16 | Aug-16 | Shahzada Ba | 49 | 104 | 37 |
| 57 | 30/08/16 | Aug-16 | Shahzada Ba | 49 | 104 | 37 |
| 57 | 01/09/16 | Sep-16 | Shahzada Ba | 64 | 217 | 57 |
| 57 | 17/09/16 | Sep-16 | Shahzada Ba | 55 | 184 | 44 |
| 57 | 08/10/16 | Oct-16 | Shahzada Ba | 48 | 250 | 58 |
| 57 | 27/10/16 | Oct-16 | Shahzada Ba | 25 | 173 | 53 |



| | A |
|---|---|
| 1 | location |
| 2 | Nizamuddin |
| 3 | Ashok Vihar |
| 4 | Shahzada Bagh |
| 5 | Shahadra |
| 6 | Janakpuri |
| 7 | Siri Fort |
| 8 | Mayapuri Indl. Area |
| 9 | Shahdara |
| 10 | BSZ Marg |
| 11 | DCE |
| 12 | N.Y. School |
| 13 | Town Hall |
| 14 | Pritampura |
| 15 | Sarojini Nagar |
| 16 | Chandni Chowk, Delhi |
| 17 | Mayapuri Industrial Area |

Main Menu

```
     stn_code        Date   Dated    location  so2  rspm  no2
0          55  13-01-2015  Jan-15  Nizamuddin   62   182   47
1          55  27-01-2015  Jan-15  Nizamuddin   54   159   46
2          55  10-02-2015  Feb-15  Nizamuddin   39   241   48
3          55  26-02-2015  Feb-15  Nizamuddin   43   210   39
4          55  10-03-2015  Mar-15  Nizamuddin   29   199   38
..        ...         ...     ...         ...  ...   ...  ...
211       531  29-10-2023  Oct-23   Pitampura   54   450   36
212       531  05-11-2023  Nov-23   Pitampura   67   161   72
213       531  27-11-2023  Nov-23   Pitampura   49   426   44
214       531  03-12-2023  Dec-23   Pitampura   64   590   56
215       531  31-12-2023  Dec-23   Pitampura   48   420   44

[216 rows x 7 columns]
                 ==============================
                    Air Polloution in Delhi
                 ==============================
                 1: Data Manipulation Menu

                 2: Data Analysis Menu

                 3: Data Visualization Menu

                 4: Exit Menu

                 ==============================
```

Manipulation Menu and sub-menus:

```
[217 rows x 7 columns]
                 Manipulation Menu
                 *****************
                 1. Insert Details of Area-wise Air Quality parameters

                 2. Delete Details

                 3. Delete a specific air quality parameter

                 4. Go back to main menu

                 Enter your choice: 1
Index(['stn_code', 'Date', 'Dated', 'location', 'so2', 'rspm', 'no2'], dtype='object')
  stn_code      Date  Dated   location  so2  rspm  no2
0     55.0 13-01-2015 Jan-15 Nizamuddin 62.0 182.0 47.0
Enter a list of values according to the columns: [45,'12-3-2012','Mar-12','anand vihar',65,143,49]
[45, '12-3-2012', 'Mar-12', 'anand vihar', 65, 143, 49]
Insertion Completed !!!!
   stn_code      Date  Dated   location  so2  rspm  no2
0      55.0 13-01-2015 Jan-15 Nizamuddin 62.0 182.0 47.0
1      55.0 27-01-2015 Jan-15 Nizamuddin 54.0 159.0 46.0
2      55.0 10-02-2015 Feb-15 Nizamuddin 39.0 241.0 48.0
3      55.0 26-02-2015 Feb-15 Nizamuddin 43.0 210.0 39.0
4      55.0 10-03-2015 Mar-15 Nizamuddin 29.0 199.0 38.0
..      ...        ...    ...        ...  ...   ...  ...
213    531.0 27-11-2023 Nov-23  Pitampura 49.0 426.0 44.0
214    531.0 03-12-2023 Dec-23  Pitampura 64.0 590.0 56.0
215    531.0 31-12-2023 Dec-23  Pitampura 48.0 420.0 44.0
216     45.0      -2003 Mar-12 anand vihar 65.0 143.0 49.0
217     45.0  12-3-2012 Mar-12 anand vihar 65.0 143.0 49.0
```

```
[218 rows x 7 columns]
                    Manipulation Menu
                    *****************
                    1. Insert Details of Area-wise Air Quality parameters

                    2. Delete Details

                    3. Delete a specific air quality parameter

                    4. Go back to main menu

                    Enter your choice: 2
   stn_code       Date   Dated    location   so2   rspm   no2
0      55.0  13-01-2015  Jan-15   Nizamuddin  62.0  182.0  47.0
1      55.0  27-01-2015  Jan-15   Nizamuddin  54.0  159.0  46.0
2      55.0  10-02-2015  Feb-15   Nizamuddin  39.0  241.0  48.0
3      55.0  26-02-2015  Feb-15   Nizamuddin  43.0  210.0  39.0
4      55.0  10-03-2015  Mar-15   Nizamuddin  29.0  199.0  38.0
..      ...        ...     ...         ...   ...    ...   ...
213    531.0  27-11-2023  Nov-23   Pitampura  49.0  426.0  44.0
214    531.0  03-12-2023  Dec-23   Pitampura  64.0  590.0  56.0
215    531.0  31-12-2023  Dec-23   Pitampura  48.0  420.0  44.0
216     45.0      -2003  Mar-12  anand vihar  65.0  143.0  49.0
217     45.0   12-3-2012  Mar-12  anand vihar  65.0  143.0  49.0

[218 rows x 7 columns]
Enter the row index you want to delete: 216
Row No. 217  is deleted
   stn_code       Date   Dated    location   so2   rspm   no2
0      55.0  13-01-2015  Jan-15   Nizamuddin  62.0  182.0  47.0
1      55.0  27-01-2015  Jan-15   Nizamuddin  54.0  159.0  46.0
2      55.0  10-02-2015  Feb-15   Nizamuddin  39.0  241.0  48.0
3      55.0  26-02-2015  Feb-15   Nizamuddin  43.0  210.0  39.0
4      55.0  10-03-2015  Mar-15   Nizamuddin  29.0  199.0  38.0
..      ...        ...     ...         ...   ...    ...   ...
212    531.0  05-11-2023  Nov-23   Pitampura  67.0  161.0  72.0
213    531.0  27-11-2023  Nov-23   Pitampura  49.0  426.0  44.0
214    531.0  03-12-2023  Dec-23   Pitampura  64.0  590.0  56.0
215    531.0  31-12-2023  Dec-23   Pitampura  48.0  420.0  44.0
217     45.0   12-3-2012  Mar-12  anand vihar  65.0  143.0  49.0
```

```
                    4. Go back to main menu

                    Enter your choice: 3
Index(['stn_code', 'Date', 'Dated', 'location', 'so2', 'rspm', 'no2'], dtype='object')
Enter the Column Name you want to delete: Dated
Are you Sure to delete(Y,N)y
Column: Dated has been deleted successfully
    stn_code       Date   location  so2  rspm  no2
0         55  13-01-2015  Nizamuddin   62   182   47
1         55  27-01-2015  Nizamuddin   54   159   46
2         55  10-02-2015  Nizamuddin   39   241   48
3         55  26-02-2015  Nizamuddin   43   210   39
4         55  10-03-2015  Nizamuddin   29   199   38
..       ...         ...         ...  ...   ...  ...
211      531  29-10-2023   Pitampura   54   450   36
212      531  05-11-2023   Pitampura   67   161   72
213      531  27-11-2023   Pitampura   49   426   44
214      531  03-12-2023   Pitampura   64   590   56
215      531  31-12-2023   Pitampura   48   420   44
```

```
*****************
1. Insert Details of Area-wise Air Quality parameters

2. Delete Details

3. Delete a specific air quality parameter

4. Go back to main menu

Enter your choice: 4
=============================
  Air Polloution in Delhi
=============================
1: Data Manipulation Menu

2: Data Analysis Menu

3: Data Visualization Menu

4: Exit Menu
```

Data Analysis Menu and sub-menus

```
-----------------------------
Enter your choice: 2
Data Frame Analysis
~~~~~~~~~~~~~~~~~~~~~~~~~
1. Top record

 2. Bottom Records

 3. To Print Specific column

 4. To Print multiple columns

 5. To display station details according to highest NO2 content

 6. To display station details according to lowest rspm parameter

 7. To Display complete statitics of the dataframe

 8. To Display complete information about dataframe
```

```
                    9.To go back
                    Enter Your Choice : 1
Enter the No. of records you want to see: 3
Top 1 records
  stn_code       Date    location  so2  rspm  no2
0       55  13-01-2015  Nizamuddin   62   182   47
1       55  27-01-2015  Nizamuddin   54   159   46
2       55  10-02-2015  Nizamuddin   39   241   48
                    Data Frame Analysis
```

```
Enter Your Choice : 2
Enter the No. of records you want to see: 4
Bottom 2 records
    stn_code       Date   location  so2  rspm  no2
212      531  05-11-2023  Pitampura   67   161   72
213      531  27-11-2023  Pitampura   49   426   44
214      531  03-12-2023  Pitampura   64   590   56
215      531  31-12-2023  Pitampura   48   420   44
                Data Frame Analysis
```

```
Enter Your Choice : 3
Name of column
 Index(['stn_code', 'Date', 'location', 'so2', 'rspm', 'no2'], dtype='object')
Enter the Column name to be displayed: stn_code
0      55
1      55
2      55
3      55
4      55
    ...
211   531
212   531
213   531
214   531
215   531
Name: stn_code, Length: 216, dtype: int64
                Data Frame Analysis
```

```
                    9.To go back
                    Enter Your Choice : 4
Name of the columns
 Index(['stn_code', 'Date', 'Dated', 'location', 'so2', 'rspm', 'no2'], dtype='object')
Enter the column names as list in square brackets[]: ['stn_code','Date','Dated']
    stn_code       Date   Dated
0         55  13-01-2015  Jan-15
1         55  27-01-2015  Jan-15
2         55  10-02-2015  Feb-15
3         55  26-02-2015  Feb-15
4         55  10-03-2015  Mar-15
..       ...         ...     ...
211      531  29-10-2023  Oct-23
212      531  05-11-2023  Nov-23
213      531  27-11-2023  Nov-23
214      531  03-12-2023  Dec-23
215      531  31-12-2023  Dec-23
```

```
                    9.To go back
                    Enter Your Choice : 5
Original Dataframe is
    stn_code       Date   Dated   location  so2  rspm  no2
0         55  13-01-2015  Jan-15  Nizamuddin   62   182   47
1         55  27-01-2015  Jan-15  Nizamuddin   54   159   46
2         55  10-02-2015  Feb-15  Nizamuddin   39   241   48
3         55  26-02-2015  Feb-15  Nizamuddin   43   210   39
4         55  10-03-2015  Mar-15  Nizamuddin   29   199   38
..       ...         ...     ...         ...  ...   ...  ...
211      531  29-10-2023  Oct-23   Pitampura   54   450   36
212      531  05-11-2023  Nov-23   Pitampura   67   161   72
213      531  27-11-2023  Nov-23   Pitampura   49   426   44
214      531  03-12-2023  Dec-23   Pitampura   64   590   56
215      531  31-12-2023  Dec-23   Pitampura   48   420   44

[216 rows x 7 columns]
station details according to highest NO2 content is
    stn_code       Date   Dated                location  so2  rspm  no2
191      345  22-12-2022  Dec-22  Mayapuri Industrial Area   45   616  223
165      146  26-11-2021  Nov-21          Chandni Chowk   68   305  207
166      146  08-12-2021  Dec-21          Chandni Chowk   75   435  182
188      345  03-11-2022  Nov-22  Mayapuri Industrial Area   33   458  171
163      146  19-10-2021  Oct-21          Chandni Chowk   40   296  162
..       ...         ...     ...                    ...  ...   ...  ...
156      146  01-07-2021  Jul-21          Chandni Chowk   43    72   29
205      531  29-07-2023  Jul-23             Pitampura   47   416   28
206      531  04-08-2023  Aug-23             Pitampura   40   171   28
180      345  01-07-2022  Jul-22  Mayapuri Industrial Area   53   164   23
133      144  29-07-2020  Jul-20          Sarojini Nagar   59    18   18
```

```
                    9.To go back
                    Enter Your Choice : 6
Original Dataframe is
   stn_code       Date   Dated    location  so2  rspm  no2
0        55  13-01-2015  Jan-15  Nizamuddin   62   182   47
1        55  27-01-2015  Jan-15  Nizamuddin   54   159   46
2        55  10-02-2015  Feb-15  Nizamuddin   39   241   48
3        55  26-02-2015  Feb-15  Nizamuddin   43   210   39
4        55  10-03-2015  Mar-15  Nizamuddin   29   199   38
..      ...         ...     ...         ...  ...   ...  ...
211     531  29-10-2023  Oct-23   Pitampura   54   450   36
212     531  05-11-2023  Nov-23   Pitampura   67   161   72
213     531  27-11-2023  Nov-23   Pitampura   49   426   44
214     531  03-12-2023  Dec-23   Pitampura   64   590   56
215     531  31-12-2023  Dec-23   Pitampura   48   420   44

[216 rows x 7 columns]
station details according to highest NO2 content is
   stn_code       Date   Dated              location  so2  rspm  no2
133     144  29-07-2020  Jul-20          Sarojini Nagar   59    18   18
137     144  23-09-2020  Sep-20          Sarojini Nagar   40    37   60
158     146  04-08-2021  Aug-21           Chandni Chowk   55    39   50
132     144  14-07-2020  Jul-20          Sarojini Nagar   69    41   49
155     146  24-06-2021  Jun-21           Chandni Chowk   64    53   62
..      ...         ...     ...                     ...  ...   ...  ...
164     146  03-11-2021  Nov-21           Chandni Chowk   45   457  155
188     345  03-11-2022  Nov-22  Mayapuri Industrial Area   33   458  171
69       58  26-11-2017  Nov-17              Shahadra   46   470   67
214     531  03-12-2023  Dec-23             Pitampura   64   590   56
191     345  22-12-2022  Dec-22  Mayapuri Industrial Area   45   616  223
```

```
                    9.To go back
                    Enter Your Choice : 7
Statistics of the database:
       stn_code        so2        rspm        no2
count  216.000000  216.000000  216.000000  216.000000
mean   161.666667   53.777778  222.634259   64.583333
std    158.316257   13.712553  106.856872   34.569546
min     55.000000   15.000000   18.000000   18.000000
25%     58.000000   45.000000  145.750000   44.000000
50%     60.000000   54.000000  206.000000   51.000000
75%    146.000000   64.250000  281.000000   71.250000
max    531.000000   94.000000  616.000000  223.000000
                    Data Frame Analysis
                    ~~~~~~~~~~~~~~~~~~~~~~~~~~
```

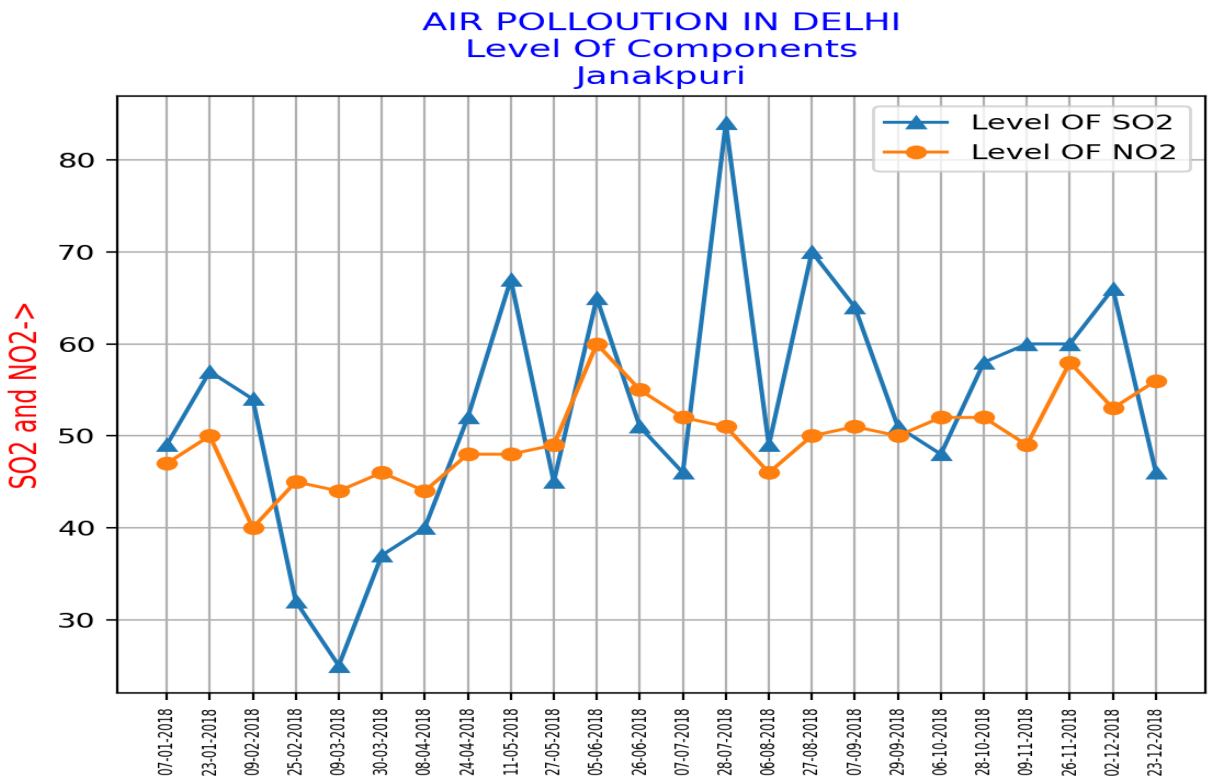
```
9.To go back
Enter Your Choice : 8
Information About The Database:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 216 entries, 0 to 215
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   stn_code  216 non-null    int64
 1   Date      216 non-null    object
 2   Dated     216 non-null    object
 3   location  216 non-null    object
 4   so2       216 non-null    int64
 5   rspm      216 non-null    int64
 6   no2       216 non-null    int64
dtypes: int64(4), object(3)
memory usage: 11.9+ KB
None
Data Frame Analysis
```

```
9.To go back
Enter Your Choice : 9
==============================
  Air Polloution in Delhi
==============================
1: Data Manipulation Menu

2: Data Analysis Menu

3: Data Visualization Menu

4: Exit Menu
```

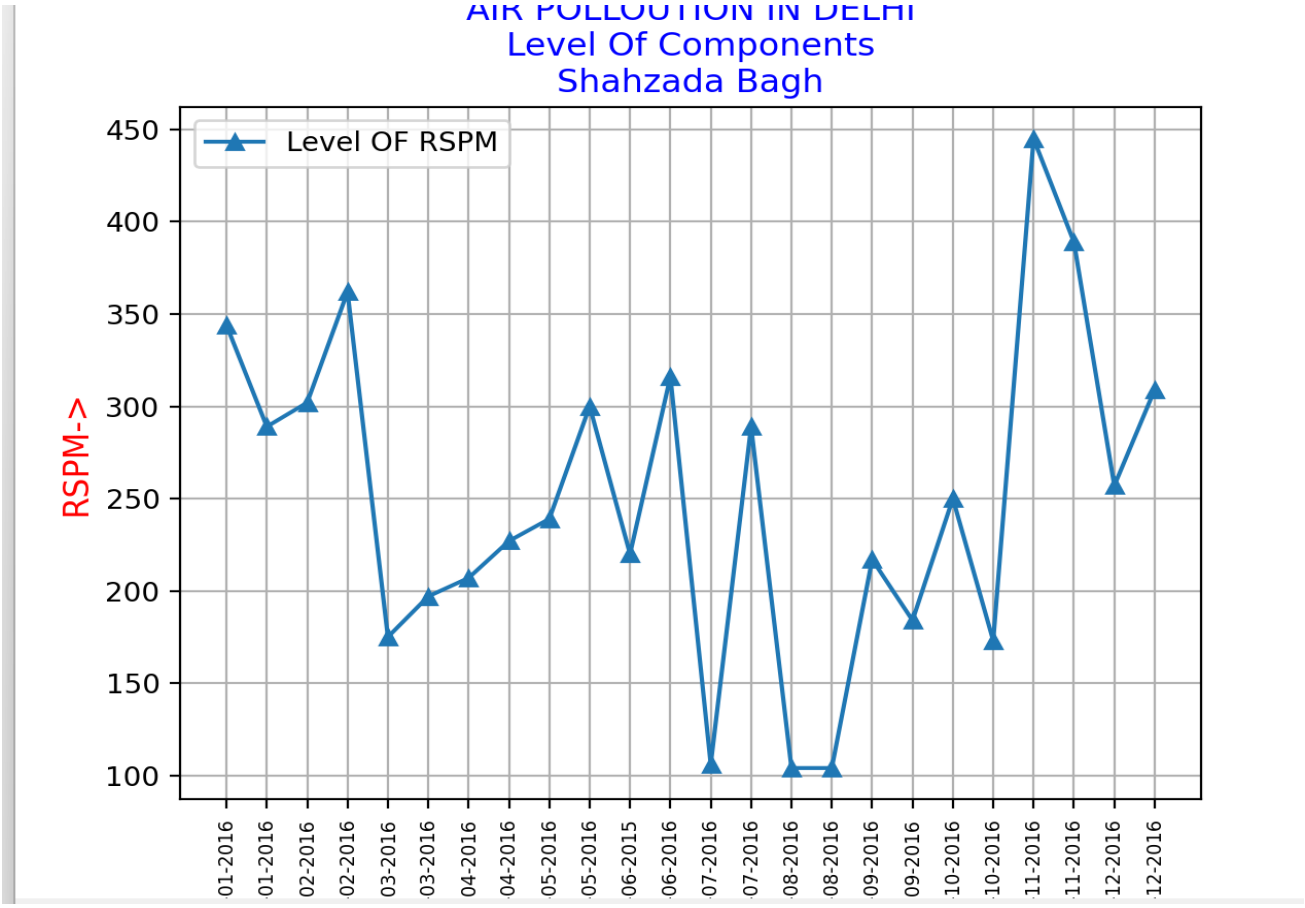
Data Visualization Menu and sub-menus:

```
-----------------------------
Enter your choice: 3

==========================
 Data Visualisation Menu
==========================
1: Line Chart Representation of SO2 and NO2 content of specific station
2: Line chart Representation of RSPM of specific location
3: Bar chart Representation of SO2 and NO2 of specific station
4: Bar Chart Representation of RSPM of specific location
5: Back
```

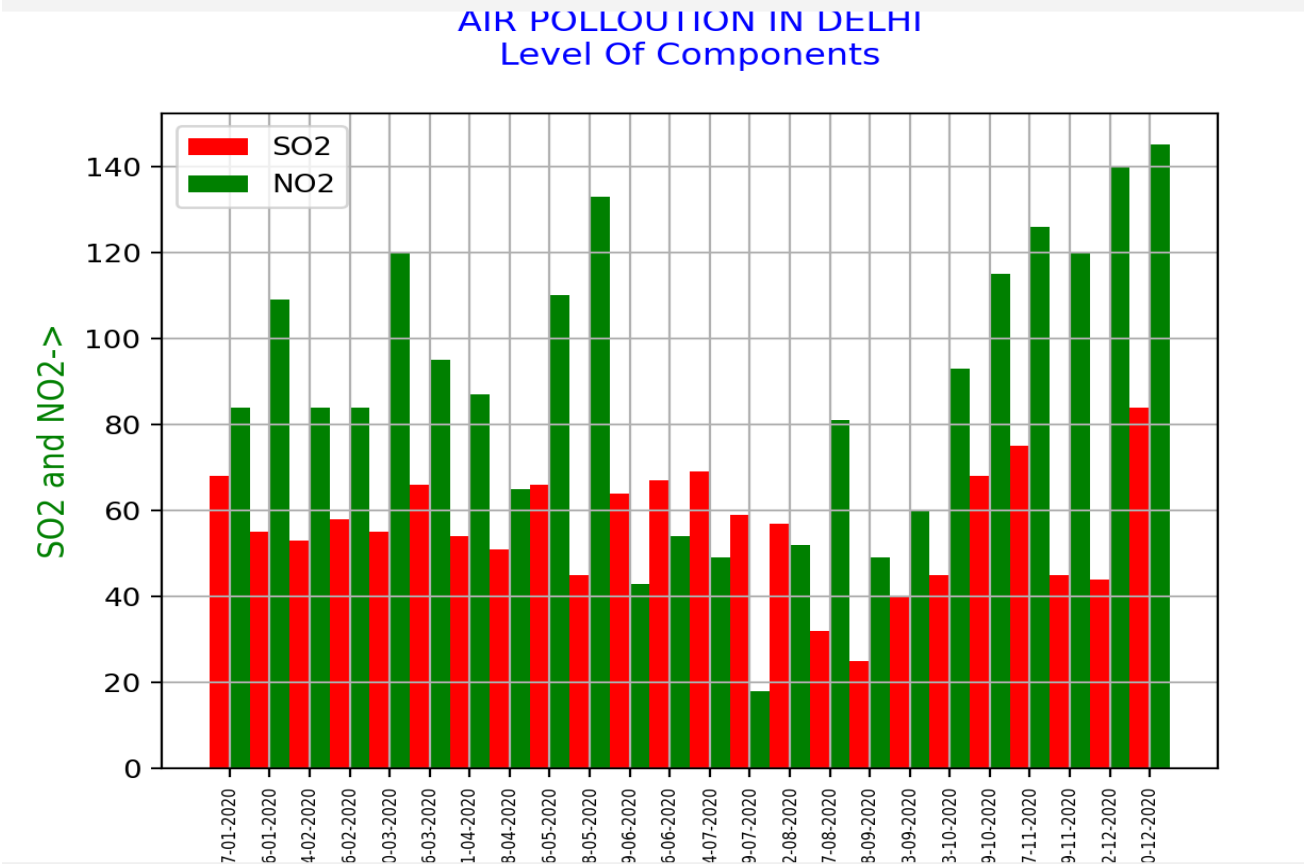
```
5: Back
Enter your choice: 1
Available Locations In New Delhi
0          Nizamuddin
1          Ashok Vihar
2          Shahzada Bagh
3          Shahadra
4          Janakpuri
5          Siri Fort
6          Mayapuri Indl. Area
7          Shahdara
8          BSZ Marg
9          DCE
10         N.Y. School
11         Town Hall
12         Pritampura
13         Sarojini Nagar
14         Chandni Chowk, Delhi
15         Mayapuri Industrial Area
Name: location, dtype: object
Enter Location Name: Janakpuri
```




```
                    ========================
                     Data Visualisation Menu
                    ========================
                    1: Line Chart Representation of SO2 and NO2 content of specific station
                    2: Line chart Representation of RSPM of specific location
                    3: Bar chart Representation of SO2 and NO2 of specific station
                    4: Bar Chart Representation of RSPM of specific location
                    5: Back
                    Enter your choice: 2
Available Locations In New Delhi
0          Nizamuddin
1          Ashok Vihar
2          Shahzada Bagh
3          Shahadra
4          Janakpuri
5          Siri Fort
6      Mayapuri Indl. Area
7          Shahdara
8          BSZ Marg
9          DCE
10         N.Y. School
11         Town Hall
12         Pritampura
13         Sarojini Nagar
14     Chandni Chowk, Delhi
15   Mayapuri Industrial Area
Name: location, dtype: object
Enter Location Name: Shahzada Bagh
```




```
                    -------------------------
                     Data Visualisation Menu
                    =========================
                    1: Line Chart Representation of SO2 and NO2 content of specific station
                    2: Line chart Representation of RSPM of specific location
                    3: Bar chart Representation of SO2 and NO2 of specific station
                    4: Bar Chart Representation of RSPM of specific location
                    5: Back
                    Enter your choice: 3
Available Locations In New Delhi
0              Nizamuddin
1             Ashok Vihar
2            Shahzada Bagh
3               Shahadra
4               Janakpuri
5               Siri Fort
6        Mayapuri Indl. Area
7               Shahdara
8               BSZ Marg
9                  DCE
10             N.Y. School
11             Town Hall
12             Pritampura
13           Sarojini Nagar
14       Chandni Chowk, Delhi
15    Mayapuri Industrial Area
Name: location, dtype: object
Enter Location Name: Sarojini Nagar
```


AIR POLLOUTION IN DELHI
Level Of Components
SO2
NO2
SO2 and NO2->
0
20
40
60
80
100
120
140
7-01-2020
6-01-2020
4-02-2020
6-02-2020
0-03-2020
6-03-2020
1-04-2020
8-04-2020
6-05-2020
8-05-2020
9-06-2020
6-06-2020
4-07-2020
9-07-2020
2-08-2020
7-08-2020
8-09-2020
3-09-2020
3-10-2020
9-10-2020
7-11-2020
9-11-2020
2-12-2020
0-12-2020

```
                    ========================
                     Data Visualisation Menu
                    ========================
                    1: Line Chart Representation of SO2 and NO2 content of specific station
                    2: Line chart Representation of RSPM of specific location
                    3: Bar chart Representation of SO2 and NO2 of specific station
                    4: Bar Chart Representation of RSPM of specific location
                    5: Back
                    Enter your choice: 4
Available Locations In New Delhi
0            Nizamuddin
1           Ashok Vihar
2          Shahzada Bagh
3              Shahadra
4              Janakpuri
5              Siri Fort
6      Mayapuri Indl. Area
7              Shahdara
8              BSZ Marg
9                 DCE
10           N.Y. School
11             Town Hall
12            Pritampura
13         Sarojini Nagar
14     Chandni Chowk, Delhi
15  Mayapuri Industrial Area
Name: location, dtype: object
Enter Location Name: Nizamuddin
```

```
                    ========================
                     Data Visualisation Menu
                    ========================
                    1: Line Chart Representation of SO2 and NO2 content of specific station
                    2: Line chart Representation of RSPM of specific location
                    3: Bar chart Representation of SO2 and NO2 of specific station
                    4: Bar Chart Representation of RSPM of specific location
                    5: Back
                    Enter your choice: 5
Going Back to Main Menu....
                    ============================
                      Air Polloution in Delhi
                    ============================
                    1: Data Manipulation Menu

                    2: Data Analysis Menu

                    3: Data Visualization Menu

                    4: Exit Menu

                    ============================
                    Enter your choice: 4
Are You Sure to exit(Y/N)y
THANK You. Closing now....
>
```

# CONCLUSION

The purpose of this project is to monitor, analyze and represent the Air Quality in various locations in Delhi through the different parameters such as $SO_2$, $NO_2$ and rspm.

This project aims to read parameter details from CSV file and then convert it in a Data Frame format. The project then performs various manipulations on data frame created such as adding parameters and records, deleting records and parameters, displaying specific records from the top and bottom, analyzing data, visualizing and representing data through bar chart and line chart. The project also displays the data according to high level of $SO_2$ and rspm.

This code can be further modified to:

- Make it compatible with a variety of languages
- Include area-wise representation of parameters
- Include comparative analysis of air pollution in various locations
- adding GUI interface for separate functionality making it user-friendly

# BIBLIOGRAPHY


1. ***https://www.geeksforgeeks.org/python-programming-language/***

2. ***https://www.kaggle.com/datasets/***

3. **"INFORMATICS PRACTICES with PYTHON" Class XII by Preeti Arora**

3. **"INFORMATICS PRACTICES Class-XII" by CBSE (NCERT)**

4. ***https://github.com/topics***